

Vineeth Gutta

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TECHNICAL SKILLS

Languages: Python, Java, C++, React JS, Node JS, Bash, SQL

Software and Tools: Git, Docker, PyTorch, TensorFlow, Scikit-Learn, Numpy, Pandas, ONNX, ONNX Runtime, REST, Eclipse, relational databases, NoSQL

High Performance Computing (HPC) software: Slurm, Singularity, Horovod, PyTorch Distributed Data Parallel, RAPIDS AI, DeepSpeed, AMD ROCm, GPU Profiling (Omnitrace, PyTorch Profiler, NSight)

EDUCATION

Doctor of Philosophy (PhD) in Computer Science (2019 – Present)
University of Delaware – Newark, DE Cumulative GPA: 3.95
Advisor: Dr. Sunita Chandrasekaran Expected: January 2025

Bachelor of Science in Computer Science (2015 – 2019)
University of Delaware – Newark, DE Cumulative GPA: 3.43

EXPERIENCE

PIConGPU Center for Accelerated Application Readiness (CAAR) Project at Oak Ridge Nat. Lab. June 2023 – Present
Computational Researcher

- PIconGPU (Particle-in-Cell) Center for Accelerated Application Readiness (CAAR): 1 of 8 projects selected in the country to stress test next-gen Dept. of Energy's Frontier Exascale system
- Ported plasma-physics DL models to the Frontier exascale system at Oak Ridge National Lab (ORNL)
- Improved scalability and optimization of ML models on AMD MI250X GPUs with the ROCm software
- Optimized application by profiling distributed training workloads using AMD profiler Omnitrace
- Ported to NVIDIA's DGX cluster (H100s) for performance and power analysis

Pacific Northwest National Lab (PNNL) Newark, DE
PhD Intern- Scalable Analytics and Decision Optimization group June 2022 – August 2022

- Created a proxy application for scalable Graph Neural Networks (GNNs)
- Created a workflow for sparse, distributed and scalable graphs in distributed memory using Louvain method for community detection

Leidos/ National Cancer Institute/ NIH Newark, DE
Computational Researcher/ Data Scientist June 2021 - Present

- Improved portability of applications like AMPL and CANDLE which leverage DL to automate key drug discovery steps and address cancer surveillance respectively
- Developed a library for assessing the predictive efficacy of cancer drug response models across Convolutional Neural Networks (CNNs) and Gradient Boosted Tree models, particularly beneficial for tabular dataset problems.
- Worked on scaling DL model training on AMD MI250X GPUs and implemented hyperparameter optimization (HPO) in parallel across many GPUs.
- Improved robustness of DL workflows built with DeepChem and RDKit for ligand-protein pose selection and binding free energy predictions.

University of Delaware – Dept. of Computer & Information Sciences Newark, DE
Graduate Teaching Assistant August 2019 - May 2021

- Assisted in several courses including Data Mining and Software Engineering
- Taught labs for Software Engineering and lead a team of undergrad and graduate TAs
- Incorporated agile methodologies such as code reviews into the curriculum

Instructor June 2020 - August 2020

- Taught the intro to object-oriented programming course
- Planned, designed, and revised syllabi, curriculum, and instruction content to enhance student learning and experience
- Achieved high ratings based on anonymous student feedback (4.57/5.0)